WEEK 3 — RESEARCH VALIDATION, HALLUCINATION & BIAS ANALYSIS

RIT 1409 PE TEAM 5 | CLIMATE CHANGE

Final consolidated submission

This document consolidates all three Week 3 requirements: Research Validation Matrix, Hallucination & Bias Detection Report, and AI-Assisted Research Report.

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Executive Summary

Week 3 evaluates the reliability of AI-generated climate-change research produced during Week 2. Three prompt outputs were checked against credible external sources. The principal findings are that the main climate-impact themes are broadly supported, while reliability depends on wording precision, source traceability, reference periods, causal distinctions and regional context. The most significant risks were overstatement, missing citation traceability, ambiguous figures and causal conflation rather than widespread fabricated information.

Deliverable 1 — Research Validation Matrix

Method: selected claims from each Week 2 AI output were checked against credible external sources. "Verified" means the claim is supported at the stated level. "Partially Verified" means the central idea is supported but wording, causality, specificity or reference period requires qualification. AI output was treated as material to validate, not as evidence.

Prompt 1 — Basic/Simple Prompt (Miriam output; Roseline validation)

#	AI claim	Verdict	Source(s)	Validation explanation
1	Warmer atmospheres hold more water: evaporation accelerates, causing prolonged, severe droughts in arid regions alongside intense, hyper-concentrated rainfall in others.	Partially Verified	IPCC AR6 WGI; WMO; HKO	Warming increases atmospheric moisture and heavy precipitation, while drought increases occur in some regions. The wording is too broad; "hyper-concentrated" and the full evaporation-to-drought chain were not established as written.
2	Elevated ocean surface temperatures fuel stronger tropical cyclones, typhoons and hurricanes with higher rainfall totals and peak wind speeds.	Partially Verified	IPCC AR6 WGI; NOAA GFDL; HKO	Evidence supports higher rainfall rates and higher peak wind speeds for the most intense tropical cyclones. It should not imply that all tropical cyclones become stronger.
3	Polar ice sheets (Greenland and Antarctica) and mountain glaciers are shrinking rapidly.	Verified	NASA; IPCC AR6 WGI	NASA and IPCC support continuing ice-sheet and glacier mass loss. "Rapidly" is qualitative but consistent with the documented trend.
4	As ocean water warms, it expands. Combined with glacial melt, global sea levels are rising, causing coastal erosion, saltwater intrusion into freshwater aquifers, and permanent loss of low-lying land.	Verified	IPCC AR6 WGI; IPCC SROCC; Met Office	Thermal expansion and land-ice loss contribute to sea-level rise and coastal impacts. Local permanence and extent of land loss require context.
5	Unprecedented rates of temperature shift force species to migrate poleward or to higher altitudes.	Partially Verified	IPCC AR6 WGII	IPCC reports about half of assessed species have shifted poleward or higher. "Unprecedented" and "force" are stronger than established evidence.
6	Species unable to adapt face extinction.	Partially Verified	IPCC AR6 WGII	Climate change increases extinction risk and some climate-driven extinctions are documented, but outcomes depend on adaptation, migration and other pressures.
7	Oceans absorb roughly 30% of emitted CO2, forming carbonic acid.	Verified	NOAA Ocean Acidification; Global Carbon Budget	NOAA states that the ocean absorbs about 30% of atmospheric CO2 released into the atmosphere. CO2

				uptake changes seawater chemistry and contributes to acidification.
8	Ocean acidification deprives shell-building organisms of carbonate ions, precipitating mass coral bleaching events and disrupting marine food webs.	Partially Verified	NOAA Ocean Acidification; NOAA Coral Reef Climate	Acidification reduces carbonate-ion availability and affects calcification. NOAA identifies marine heat stress as a direct driver of coral bleaching; the original wording mixes mechanisms.
9	Glacial retreat reduces seasonal runoff for rivers that supply fresh water to hundreds of millions of people, while droughts strain municipal reservoirs.	Partially Verified	IPCC SROCC; UNU	Glacier runoff can rise initially and later decline after peak water. The exact "hundreds of millions" figure and reservoir statement were not verified as worded.
10	Destruction of livelihoods, persistent crop failures and flooding create millions of "climate refugees" or internally displaced persons, heightening geopolitical tensions over resources.	Partially Verified	IPCC AR6 WGII; IDMC; UNHCR	Climate hazards contribute to displacement. "Climate refugee" is not a general legal status, most displacement is internal, and the geopolitical-causation link was not established.

Prompt 2 — Structured RCTFC Prompt (Miriam finalized validation)

#	AI claim	Verdict	Source(s)	Validation explanation
1	Climate change affects ecosystems, biodiversity, water resources, food systems, and human societies.	Verified	IPCC AR6 WGII	The IPCC assesses impacts across ecosystems, food and water security and human systems.
2	Rising temperatures and changing rainfall patterns can increase droughts, floods, and other extreme weather events.	Verified	IPCC AR6 WGII	The IPCC documents changes in climate extremes and links warming to changes in heavy precipitation and drought risk in relevant regions.
3	These changes can damage ecosystems and reduce the ability of some species to survive or adapt.	Verified	IPCC AR6 WGII	The IPCC documents ecosystem degradation, species range shifts and increased risks where species cannot adapt or move quickly enough.
4	Climate change can affect homes, livelihoods, access to water and food, and living conditions.	Verified	WHO; IPCC AR6 WGII	WHO and IPCC document effects on food/water security, livelihoods, settlements and living conditions.
5	Floods, droughts, storms, and other climate-related hazards can cause displacement and place pressure on communities and essential services.	Verified	IPCC AR6 WGII; World Bank material cited by team	Climate hazards can contribute to displacement and strain services. Displacement also has non-climate drivers.
6	Climate change can cause economic losses by affecting agriculture, fisheries, tourism, infrastructure, and people's ability to work.	Verified	World Bank material; IPCC	The team sources document losses across climate-sensitive sectors and labour productivity. Exact monetary figures need separate sourcing.
7	Extreme weather events can damage infrastructure and disrupt economic activities.	Verified	World Bank/UNFCCC case material	The team cited documented infrastructure and economic disruption, including the Manila case.
8	Climate change affects human health through extreme heat, floods, storms, food insecurity, and changes in some infectious and vector-borne diseases.	Verified	WHO Climate Change and Health	WHO identifies these direct and indirect health pathways.
9	Climate-related disasters can also affect mental health and disrupt health services.	Verified	WHO; OHCHR/PAHO material	The cited sources support mental-health impacts and pressure on health systems.
10	The severity varies across regions and communities.	Verified, broad	IPCC AR6 WGII; WHO	Regional differences in vulnerability and exposure are well established, but the

				statement itself is low in specificity.

Prompt 3 — Detailed/Controlled RCTFC Prompt (Chiamaka validation refined for final compilation)

#	AI claim	Verdict	Source(s)	Validation explanation
1	Global mean surface temperatures increased by approximately 1.1–1.2°C above pre-industrial levels between 1850–1900.	Verified	IPCC AR6; NOAA indicators cited by team	The sources support approximately 1.1–1.2°C relative to the 1850–1900 baseline; the comparison period should be stated precisely.
2	Atmospheric CO2 exceeds 50% above pre-industrial level.	Verified, reference-period qualification	NOAA; IPCC AR6; NASA indicator cited by team	The team sources support crossing the 50% threshold in recent observations, but different baselines and years must be stated.
3	Approximately 3.3–3.6 billion people live in contexts highly vulnerable to climate change.	Verified	IPCC AR6 WGII	The IPCC reports billions living in highly vulnerable contexts. The assessment context and date should accompany the figure.
4	Approximately half of assessed species globally have shifted their geographic ranges poleward or to higher elevations; ocean acidification and marine heatwaves have driven widespread coral bleaching and severe habitat degradation.	Partially Verified	IPCC AR6 WGII; NOAA	The species-range statement is supported. Acidification affects calcification, while marine heat stress is a major direct driver of bleaching; mechanisms should be separated.
5	Global mean sea level has risen by approximately 20–24 cm since 1880, escalating coastal erosion and tidal flooding.	Verified, source-period qualification	NOAA; IPCC AR6 WGI	NOAA supports roughly 21–24 cm since 1880; IPCC gives about 20 cm over 1901–2018. Different reference periods explain the difference.
6	Warmer air temperatures accelerate evaporation and hold more moisture, compounding both intense drought conditions in arid regions and heavy precipitation events elsewhere.	Verified, regional qualification	IPCC AR6 WGI	The IPCC supports hydrological-cycle intensification, including heavier precipitation and drought in some regions.
7	Food security shifts on precipitation and extreme heat undermine crop yield and marine fisheries productivity, threatening food access in climate-sensitive regions.	Verified, wording refinement	IPCC AR6 WGII	The IPCC documents reduced food/water security, slowed agricultural productivity growth and adverse effects on fisheries/shellfish in some regions.
8	Low-emission societies often suffer the most severe impacts, worsening pre-existing social inequalities, gender disparities and regional instabilities.	Partially Verified	IPCC AR6 WGII	The IPCC supports disproportionate vulnerability and gender-related differences, but “low-emission societies” and “regional instabilities” need more precise evidence.
9	Extreme weather events such as intense hurricanes, wildfires and riverine floods cause billions of dollars in damage to transport, energy and residential infrastructure.	Verified, scope qualification	IPCC; NOAA NCEI; team sources	Major infrastructure losses are supported. Monetary totals should identify geography, event and period.
10	Long-term risk projections rely on Shared Socioeconomic Pathways (SSPs); future impacts depend on policy choices, technological development and emissions trajectories that cannot be predicted with	Verified	IPCC AR6	IPCC scenario-based projections are conditional on socioeconomic, technology, emissions and policy pathways.

	total certainty.			

Consolidated Deliverable 1 Findings

Prompt 1: 3 selected claims were Verified and 7 Partially Verified, with none classified Unsupported/Unverified. Prompt 2 was generally supported but comparatively general. Prompt 3 contained detailed quantitative and scientific claims that were largely supportable but required careful treatment of baselines, reference periods and causal mechanisms. Verification does not eliminate hallucination risk; it reduces risk when claims are independently checked and qualified.

Deliverable 2 — Hallucination & Bias Detection Report

1. Purpose

This report examines hallucinations, unsupported or overstated claims, citation limitations, potential bias/framing and methodological limitations across the three Week 2 climate-change outputs.

2. Hallucination Findings

- Fabricated sources/events: No fabricated source, quotation, study or event was identified in the selected claims. The main issue was traceability because Prompt 1 had no inline citations.
- Unsupported wording: Details such as “hyper-concentrated rainfall,” “hundreds of millions,” and the geopolitical-tension conclusion were not established as written.
- Overgeneralization: Some supported ideas were expressed universally, such as all tropical cyclones becoming stronger or species unable to adapt facing extinction.
- Causal conflation: Prompt 1 connected ocean acidification directly to mass coral bleaching. NOAA distinguishes ocean acidification from heat-stress-driven bleaching.
- Reference-period ambiguity: Different valid figures can result from different reference periods. Prompt 3 sea-level figures must be accompanied by their periods/baselines.

3. Bias and Framing Analysis

No evidence of intentional bias was established. The concern is framing: short AI answers can make climate impacts appear uniform across countries and communities. IPCC and WHO evidence shows that vulnerability, exposure, adaptive capacity, infrastructure and health-system conditions differ substantially.

- Uniformity: use regional and contextual qualifiers.
- Excess certainty: replace universal wording with evidence-matched language.
- Selective mechanism: separate ocean acidification effects from thermal-stress-driven bleaching.
- Untraceable evidence: attach sources, dates and reference periods to quantitative claims.
- Terminology: prefer “climate-related displacement” rather than treating “climate refugee” as a universal legal category.

4. Prompt-by-Prompt Critical Evaluation

Prompt 1: useful breadth but the least controlled wording; main weaknesses were absent citations, overstatement, ambiguous quantitative language and limited context.

Prompt 2: clearer four-category structure and mostly general, supportable claims; its limitation is low evidentiary specificity.

Prompt 3: explicit anti-fabrication and uncertainty instructions improved research-style structure, but detailed outputs still required claim-level verification.

5. Recommendations

- Require source attribution for quantitative claims.
- State reference periods and baselines.
- Separate observed facts, projections and interpretations.
- Use regional qualifiers where evidence varies.
- Check exact wording rather than only the general idea.

- Separate distinct causal mechanisms.
- Require human verification before final submission.
- Treat anti-hallucination instructions as risk reduction, not guarantees.

6. Limitations

This is a structured validation exercise, not a systematic literature review. Only ten selected claims per prompt were examined. Some sources are institutional summaries or agency pages. Figures can change as datasets are updated. The review does not establish that every unselected sentence in the Week 2 outputs is correct.

7. Conclusion

The principal reliability problems were overstatement, missing source traceability, ambiguous figures, causal conflation and insufficient context rather than widespread fabricated information. Responsible AI-assisted research requires validation of the exact claim.

Deliverable 3 — AI-Assisted Research Report

Major Impacts of Climate Change on Humans and the Environment

Introduction

Climate change affects natural ecosystems and human societies through changes in temperature, precipitation, extreme events, sea level and ocean chemistry. The IPCC assesses widespread adverse impacts on nature and people, including ecosystems, food and water security, settlements, infrastructure and health.

Research Methodology

Three prompt designs were tested in Week 2: basic, structured RCTFC and detailed/controlled RCTFC. Week 3 selected important claims and checked them against credible sources, focusing on statistics, baselines, reference periods, causal language and regional qualifiers.

Validated Environmental Findings

Evidence supports ecosystem and biodiversity impacts, cryosphere loss, sea-level rise and ocean-chemistry changes. NOAA explains that ocean acidification reduces carbonate-ion availability; NOAA separately identifies warming-related thermal stress as a driver of coral bleaching.

Food and Water Security

The IPCC reports reduced food and water security and slowed global agricultural productivity growth, with regional variation. Glacier retreat can alter downstream water availability over time, depending on basin and glacier conditions.

Health and Social Impacts

WHO and IPCC evidence supports increased risks from heat and other climate-related hazards, including physical and mental health effects. Climate hazards can contribute to displacement and pressure on services, but such outcomes have multiple drivers.

Economic and Infrastructure Impacts

Climate-related extremes can damage infrastructure and climate-sensitive sectors and affect labour productivity. Monetary figures require event, geography, period and source.

Vulnerability and Inequality

The IPCC finds that vulnerable people and systems are disproportionately affected. Exposure, poverty, infrastructure, health services, livelihood dependence and adaptive capacity shape outcomes.

Responsible AI-Assisted Research Procedure

Generate a structured draft; extract factual claims; check claims against credible sources; verify wording, numbers, dates, baselines and periods; mark partially supported claims; separate evidence from interpretation/projection; document AI assistance; retain human responsibility for final accuracy.

Conclusion

AI can assist with organizing climate-change research, but reliable research requires claim-level verification. The final integrated findings are limited to validated or appropriately qualified claims.

Transparent AI-Use Statement

AI tools were used during the Week 2 prompt experiment and during preparation and organization of Week 3 validation materials. Claude was used in Roseline's documented process to help locate sources and draft validation text. AI output was treated as material to be checked, not as evidence. Final responsibility for accuracy, source selection and submission remains with the human research team.

References

IPCC (2022), Climate Change 2022: Impacts, Adaptation and Vulnerability — Summary for Policymakers
https://www.ipcc.ch/report/ar6/wg2/downloads/report/IPCC_AR6_WGII_SummaryForPolicymakers.pdf

IPCC (2021), Climate Change 2021: The Physical Science Basis
https://www.ipcc.ch/report/ar6/wg1/

NOAA, Ocean Acidification
https://oceanservice.noaa.gov/facts/acidification.html

NOAA, How does climate change affect coral reefs?
https://oceanservice.noaa.gov/facts/coralreef-climate.html

WHO, Climate change and health
https://www.who.int/news-room/fact-sheets/detail/climate-change-and-health

WHO, Heat and health
https://www.who.int/news-room/fact-sheets/detail/climate-change-heat-and-health

NASA Science, Ice Sheets
https://science.nasa.gov/earth/explore/earth-indicators/ice-sheets

IDMC, Global Report on Internal Displacement
https://www.internal-displacement.org/global-report/

Hong Kong Observatory, Climate Change
https://www.hko.gov.hk/en/climate_change/

Final Quality-Control Checklist

- Deliverable 1 includes all three prompt validation matrices.
- Each prompt contains 10 selected claims with verdicts, sources and explanations.
- Chiamaka's hallucination-risk wording and causal/reference-period issues are corrected in the consolidated interpretation.
- Deliverable 2 covers hallucination, unsupported claims, citation issues, bias/framing and limitations.
- Deliverable 3 integrates validated findings into a professional research report.
- AI use is disclosed and human verification is stated.
- References are included.
- The Week 3 submission does not depend on any unreceived teammate contribution.